YARN WORKING .pdf

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Summary

This text serves as a conceptual guide to how **YARN manages and executes distributed computing jobs** by using a WordCount program as a practical demonstration. The workflow begins when a user submits an application, which is then broken down into individual **tasks that run within isolated resource units called containers**. To ensure efficiency, a central **Resource Manager** coordinates with an **Application Master** to distribute these tasks across various nodes, prioritizing **data locality** to minimize network congestion. Ultimately, the source clarifies the distinct responsibilities of each architectural component, illustrating how they collaborate to **allocate hardware resources and monitor task completion** across a cluster.

Keywords: YARN Job Allocation, Resource Manager, Application Master, Container Management, Data Locality

Content

How YARN Allocates Jobs – Explained with Example
Understand job (application) allocation in YARN with a simple WordCount example
1 Key Terms
Job (Application)
A program submitted by the user Words.a.Cotint, Log Analysis).
In YARN, this is called an Application.
Task
A job is divided into tasks.
Example (WordCount): 4 Map tasks |+1| Reduce task *c* = 5 tasks total
Container
A resource unit (CPU + Memory) where a task runs.
Example: 2 GB RAM + 1 CPU
3 User Submits a Job
WordCount Job Submitted

Input file size $= 1\ GB$
↓
HDFS splits file into 4 blocks
Block1
Block2
Block3
↓
Block4
Example Cluster
Node1
(Node Manager)
4 GB RAM
2 Containers
Resource Manager
Node2
(Node Manager)
4 GB RAM
2 Containers
Node3
(Node Manager)
Total containers in cluster $= 6$
4 Step 1: Resource Manager Starts
Application Master
Node1
Container 1
Application
Master
Container 2
Free
Node2
Container 1
Free
Container 2
Free
4 GB RAM
2 Containers
Node3
Container 1
Free
Container 2
Free

MapReduce creates tasks:
4 Map tasks (one for each block)
1 Reduce task
Total tasks
 = 5
Step 2: Application Master
Requests Containers
Application Master asks Resource Manager:
“I need 5 containers for
4 Map tasks and 1 Reduce task”
The first container is used to launch the Application Master.
6 Step 3: Resource Manager Allocates Containers
Scheduler distributes 5 containers across the nodes:
Node1
Container 1
Application
Master
Node2
Node3
Container 1
Map Task 2
Container 1
Map Task 4
Container 2
Map Task 1
Container 2
Map Task 3
Container 2
Reduce Task
Map Tasks Complete
Step 4: Node Managers Start Tasks
Node Managers launch the containers
and run the tasks.
Each node reports status to the
Application Master.
8 Step 5: Tasks Finish
Data Locality (Important)
YARN tries to run tasks on the nodes where data blocks are stored to reduce network traffic.
So Map tasks are preferably
scheduled as:

Node1 Map Task for Block1
Node2 Map Task for Block2
Example block placement:
Node2 Map Task for Block3
Node3 Map Task for Block4
Shuffle & Sort
↓
Who Does What?
Reduce Task Runs
Component
Responsibility
Client
Submits the job (application)
Final Flow
Client
Resource Manager
↓
↓
Resource Manager
Allocates containers to applications
Application Master
Output written to HDFS
Application Master
Requests containers and monitors
job progress
Resource Manager
allocates containers
Node Managers start
Node Manager
Starts and stops containers, runs tasks
Container
Executes a single task
containers
Tasks execute